

UNIT- I

HUMAN ASSIST DEVICE

Human Assist Devices are devices specifically designed to support, enhance, or restore the physical, sensory, cognitive, or communicative abilities of individuals, particularly those with disabilities or impairments, to improve their independence and quality of life.

Emergency assist devices are devices used to help people during medical emergencies. When someone's health suddenly becomes dangerous—like when they stop breathing, have a heart attack, or get badly injured—these devices provide immediate help. Their main job is to save lives, keep the patient stable until full treatment is available, and make emergency care faster and more effective.

Percutaneous mechanical circulatory support

Percutaneous mechanical circulatory support (MCS) refers to a range of medical devices used to support the heart's function in patients experiencing severe cardiac dysfunction, particularly those with conditions like heart failure, cardiogenic shock, or during high-risk procedures such as percutaneous coronary interventions (PCI). These devices are inserted through the skin, typically via the femoral artery, and provide temporary circulatory support to maintain blood flow while the heart recovers or while awaiting other interventions like a heart transplant.

- Percutaneous Mechanical Circulatory Support (MCS): Temporary devices inserted through the skin (usually via arteries) to assist the heart in pumping blood.
- Purpose:
 - Supports failing heart function.

- Maintains blood flow to vital organs.

What is Mechanical Circulatory Support?

- Devices help by:
 - Reducing the workload of the heart.
 - Ensuring blood is pumped effectively to vital organs.

Types of Percutaneous Mechanical Circulatory Support (pMCS) devices

1. Intra-Aortic Balloon Pump (IABP)

A catheter with a balloon is inserted into the **descending aorta** (the large artery carrying blood from the heart to the body). The balloon inflates during **diastole** (when the heart relaxes) to push blood back toward the coronary arteries, improving oxygen supply to the heart. It deflates during **systole** (when the heart contracts) to reduce pressure in the aorta, lowering the heart's workload.

Advantages:

- Simple and quick to insert.
- Improves blood flow to the heart while reducing its oxygen demand.
- Often used in **cardiogenic shock** or after heart attacks.

2. Impella Device

A small, catheter-based microaxial pump is inserted through the femoral artery (or axillary artery for longer-term use) and positioned across the aortic valve. It directly pulls blood from the left ventricle and pumps it into the aorta, effectively reducing the heart's workload.

Advantages:

- Provides stronger circulatory support than an IABP.
- Directly unloads the heart, helping it recover faster.

3. TandemHeart

A catheter is inserted into the **left atrium** through the **femoral vein** and positioned via a small puncture in the heart's wall (called a **transseptal**

puncture). Blood is drawn from the left atrium, oxygenated externally, and returned to the body through the **femoral artery**.

Advantages:

- Provides strong circulatory support (up to **5 L/min** of blood flow).
- Useful in severe heart failure cases when other devices may not be sufficient.

4. Veno-Arterial Extracorporeal Membrane Oxygenation (VA-ECMO)

VA-ECMO draws oxygen-poor blood from a **large vein** (like the femoral vein), oxygenates it using an **external oxygenator**, and returns the oxygen-rich blood to a **major artery** (often the femoral artery or axillary artery). This supports both the **heart** and **lungs** in critical conditions.

Advantages:

- Provides **full cardiopulmonary support** in cases of severe heart or lung failure.
- Effective in life-threatening conditions like **cardiac arrest**, **septic shock**, or **post-surgical heart failure**.

Indications for MCS

- Cardiogenic shock (e.g., after a heart attack).
- Severe heart failure (acute or chronic).
- High-risk percutaneous coronary interventions (PCI).
- Bridge-to-recovery (supporting the heart while it heals).
- Bridge-to-transplant (waiting for a heart transplant).

Applications of MCS

- Emergency use: Cardiogenic shock after a heart attack.
- During high-risk PCI: Supporting the heart during interventions.

- Bridge-to-recovery: While the heart heals.
- Bridge-to-transplant: Waiting for a heart transplant.

Ventricular Assist Devices (VADs)

- VADs are mechanical circulatory support devices used to assist the pumping function of the ventricles in patients with severe heart failure.
- Ventricular Assist Devices are mechanical devices used to support the function of failing ventricles in the heart. They are particularly useful for patients with advanced heart failure who have reduced cardiac output.
- Purpose:
They help improve blood flow in individuals whose hearts cannot pump effectively due to advanced heart disease or after a heart attack.

Classification of VADs

Based on Ventricular Support

1. Left Ventricular Assist Device (LVAD):
 - Assists the left ventricle in pumping blood into the aorta.
 - Commonly used, as the left ventricle is more prone to failure.
2. Right Ventricular Assist Device (RVAD):
 - Assists the right ventricle in pumping blood to the pulmonary artery.
 - Used in conditions like right heart failure or pulmonary hypertension.
3. Biventricular Assist Device (BiVAD):
 - Supports both ventricles.
 - Used in severe biventricular failure.

Based on duration of support

1. Short-Term VADs

- Used for days to weeks. Examples: ECMO (Extracorporeal Membrane Oxygenation), Impella devices.

2. Long-Term VADs

- Designed for months to years. Examples: HeartMate III, HeartWare HVAD.

Components of a VAD

1. **Pump:** Drives blood from the ventricle to the aorta or pulmonary artery.
2. **Cannulae:** Tubes to connect the pump with the heart and blood vessels.
3. **Controller:** External device to control the pump's speed and monitor its function.
4. **Power Source:** Batteries or external power supply.

Working Principle

1. Blood is drawn from the failing ventricle into the device using cannulae.
2. The pump propels the blood into the aorta or pulmonary artery.
3. The flow rate and pressure are controlled by the controller.

Applications of VADs

1. Bridge to Transplantation (BTT): Supports patients awaiting a heart transplant.
2. Bridge to Recovery (BTR): Temporarily supports the heart while it recovers.
3. Destination Therapy (DT): Permanent support for patients ineligible for heart transplantation.

Advantages of VADs

- Restores normal blood flow.
- Improves organ function.

- Enhances quality of life in patients with end-stage heart failure.

Biventricular Assist Devices (Bi-VADs)

Mechanical devices used to support the function of one or both failing heart ventricles. Biventricular Assist Device (Bi-VAD): Supports both the left and right ventricles.

- Purpose: To assist in pumping blood from both ventricles when the heart is unable to function effectively.
- Indication: Used in patients with severe biventricular heart failure (e.g., dilated cardiomyopathy, myocarditis, or advanced heart failure).

Components of a BiVAD

1. Pumps:
 - Two independent pumps support the left and right ventricles.
 - Some systems use a single pump with dual chambers to serve both sides.
2. Cannulae:
 - Flexible tubes that connect the BiVAD to the heart and blood vessels.
 - Blood is drained from the ventricles and returned to the aorta (left side) and pulmonary artery (right side).
3. Controller:
 - A computerized system monitors and regulates the pump's function.
 - Displays parameters such as pump flow rate, pressures, and alarm

Working Mechanism

- **Left Ventricular Support:**

- Blood is drawn from the left ventricle or left atrium and pumped into the aorta.
- Restores blood flow to vital organs.
- **Right Ventricular Support:**
 - Blood is drawn from the right ventricle or right atrium and pumped into the pulmonary artery.
 - Facilitates oxygenation in the lungs.
- **Synchronization:**
 - BiVADs work in coordination with the patient's native heart or completely take over the pumping function if the heart is not working.

Indications for Bi-VAD Use

1. Bi-VADs are used in patients with severe biventricular failure due to various underlying conditions:
2. Dilated Cardiomyopathy: Weakening of the heart muscle causing inefficient pumping.
3. Myocarditis: Inflammation of the heart muscle leading to reduced contractility.
4. Advanced Congenital Heart Diseases: Complex structural defects involving both ventricles.
5. Post-Cardiac Surgery Complications: Failure of the heart to recover postoperatively.
6. Acute Myocardial Infarction (Heart Attack): Leading to extensive damage to both ventricles.

Types of BiVADs

1. **Paracorporeal Devices:** Part of the device remains outside the body (e.g., Thoratec PVAD).

2. **Implantable Devices:** Fully implanted within the body (e.g., Syncardia Total Artificial Heart when used as a BiVAD).

Advantages

- Improves survival rates in advanced heart failure.
- Enhances quality of life by restoring cardiac output.
- Serves as a life-saving option for transplant candidates.

Cardiac Extracorporeal Membrane Oxygenation (ECMO)

ECMO is a form of advanced life support that temporarily takes over the function of the heart and/or lungs in patients with severe cardiac or respiratory failure. It is used when conventional therapies (like medications, ventilators, or intra-aortic balloon pumps) fail to maintain adequate oxygenation and circulation. ECMO provides a way to oxygenate blood and circulate it in the body while bypassing the patient's heart and/or lungs. It helps reduce the workload on these organs, allowing them to heal.

Types of ECMO

1. Veno-Arterial (VA) ECMO:
 - Supports both heart and lungs.
 - Used in cases of severe cardiac failure, cardiac arrest, or cardiogenic shock.
 - Blood is drawn from a large vein (e.g., femoral or internal jugular vein) and returned to a major artery (e.g., femoral or aortic artery).
2. Veno-Venous (VV) ECMO:
 - Supports only the lungs.

- Used in respiratory failure (e.g., ARDS or severe pneumonia).
- Blood is drawn from and returned to the venous system, bypassing the lungs.

Components of ECMO

1. Cannulae:

- Flexible tubes inserted into blood vessels to allow blood flow in and out of the ECMO circuit.
- VA ECMO: One cannula in a vein for drainage and another in an artery for return.
- VV ECMO: Both cannulae are placed in veins.

2. Pump:

- Centrifugal or roller pump creates blood flow through the ECMO circuit, simulating the heart's pumping action.

3. Oxygenator:

- Also known as an artificial lung.
- Removes carbon dioxide and adds oxygen to the blood.

4. Heat Exchanger:

- Regulates blood temperature before returning it to the patient.

5. Monitor and Control Systems:

- Sensors track flow rate, oxygenation, and pressure to ensure optimal functioning of the system.

Supportive Equipment

1. Anticoagulation monitoring (e.g., activated clotting time or thromboelastography).
2. Bedside imaging for cannula placement verification.

How ECMO Works

The Process

1. Cannulation

- Large blood vessels (femoral, jugular, or subclavian) are accessed.
- Blood is drawn from the patient (via venous cannula) and enters the ECMO circuit.

2. Oxygenation and CO₂ Removal

- Blood flows through the oxygenator, where oxygen is added, and CO₂ is removed.

3. Blood Return

- Oxygenated blood is returned to the body through the arterial or venous cannula, depending on the ECMO type.

Key Monitoring Parameters

- Blood oxygenation levels (SpO₂, pO₂).
- CO₂ clearance (pCO₂).
- Hemodynamics (blood pressure, cardiac output).
- Cannula flow rates.

When to Use ECMO

1.Cardiac-related indications

- Cardiogenic shock.
- Acute myocardial infarction with severe pump failure.
- Post-cardiotomy low cardiac output syndrome.
- Myocarditis or cardiomyopathy.
- Cardiac arrest (as part of ECPR: Extracorporeal Cardiopulmonary Resuscitation).
- Bridge to recovery, heart transplant, or a VAD.

2.Respiratory-related indications: When VV ECMO is used, such as in ARDS.

Clinical Applications in Cardiac Care

Example Cases:

1. Acute Myocardial Infarction: ECMO is used to stabilize the patient while definitive treatment (e.g., coronary intervention) is performed.
2. Cardiac Arrest: ECPR can provide circulation during resuscitation efforts.
3. Bridge to Transplant: Patients awaiting a heart transplant can be maintained on ECMO.

Duration of ECMO Support:

- ECMO is a temporary measure, usually used for days to weeks, depending on the patient's condition and response.

Complications of ECMO

1. Bleeding:
 - Due to anticoagulation therapy to prevent clotting in the circuit.
2. Thrombosis:
 - Clots forming in the circuit or cannula.
3. Infections:
 - From cannulation sites or long-term ECMO use.
4. Neurological Complications:
 - Stroke or seizures.
5. Limb Ischemia:
 - Compromised blood flow to the limb due to arterial cannulation.

Recent Advances in ECMO

1. Portable ECMO Devices:

- Allow for emergency use in ambulances or during patient transport.

2. Miniaturized Systems:

- Reduced size and complexity for easier use.

3. Artificial Intelligence:

- Real-time monitoring and predictive algorithms for managing ECMO complications.

Heart-Lung Machine

The Heart-Lung Machine (HLM), also known as a cardiopulmonary bypass machine, is a medical device that temporarily takes over the function of the heart and lungs during cardiac surgery. It allows surgeons to operate on a still and bloodless heart by maintaining oxygenation, blood circulation, and homeostasis during procedures such as heart valve replacement, coronary artery bypass grafting (CABG), or heart transplants.

principle

The machine functions by diverting blood from the patient's body, oxygenating it outside the body, and then returning it to the patient. This bypasses the heart and lungs and ensures the proper circulation and oxygenation of tissues during surgery.

Components of the Heart-Lung Machine

1. Oxygenator:

- Mimics lung function by facilitating gas exchange.
- Blood flows through this device, where oxygen is added, and carbon dioxide is removed.

- Types:
 - Bubble Oxygenators: Use bubbles to mix oxygen directly with blood (obsolete due to complications).
 - Membrane Oxygenators: Use a thin membrane to separate blood from oxygen, allowing safer gas exchange.
- 2. Pump:
 - Mimics the heart's function by ensuring continuous blood flow.
 - Types:
 - Roller Pumps: Squeeze blood through tubes by mechanical rollers.
 - Centrifugal Pumps: Use centrifugal force to propel blood.
- 3. Heat Exchanger:
 - Maintains the patient's body temperature during surgery.
 - Can cool the blood to reduce metabolic rate or warm it up before bypass termination.
- 4. Cannulae:
 - Tubes inserted into large blood vessels or the heart.
 - Venous Cannulae: Drain deoxygenated blood from the body.
 - Arterial Cannulae: Return oxygenated blood to the body.
- 5. Filters:
 - Prevent air bubbles (de-airing) and remove particulates from the blood.
- 6. Reservoir:
 - Collects blood from the body before it enters the oxygenator.
 - Types: Open or closed reservoirs.

Principle of Operation

The machine operates by diverting blood flow through an extracorporeal circuit:

1. Venous Drainage: Deoxygenated blood is drained from the patient through the venous cannula.
2. Gas Exchange: Blood flows through the oxygenator, where oxygen is added, and carbon dioxide is removed.
3. Temperature Regulation: Blood passes through the heat exchanger to adjust its temperature.
4. Reoxygenated Blood Return: The oxygenated blood is pumped back into the body via the arterial cannula.

Steps in Using the Heart-Lung Machine

1. Cannulation:
 - Surgeon places cannulae in appropriate blood vessels (e.g., superior vena cava, aorta).
2. Initiation of Bypass:
 - Blood flow is gradually diverted to the machine.
 - The heart is isolated, and circulation is maintained by the machine.
3. Monitoring:
 - Real-time monitoring of blood flow, oxygenation levels, pressure, and temperature.
4. Surgical Intervention:
 - With the heart stopped and the blood diverted, the surgeon can operate on a still, bloodless field.
5. Weaning Off Bypass:
 - Blood flow is gradually returned to the heart, and normal heart function is restored.
6. Decannulation:
 - Cannulae are removed, and the bypass circuit is dismantled.

6. Applications

- Open-heart surgeries: Coronary artery bypass grafting (CABG), valve repair/replacement.
- Correction of congenital heart defects: Atrial septal defect, ventricular septal defect.
- Complex aortic surgeries: Aneurysm repair.
- Support in extracorporeal membrane oxygenation (ECMO): Prolonged heart and lung support.

Advantages of Heart-Lung Machine

- Maintains Circulatory Function: It maintains oxygenation and blood circulation during surgery, which would otherwise be impossible with an arrest of the heart.
- Enables Complex Cardiac Surgeries: Essential for surgeries that involve direct manipulation of the heart.
- Temperature Control: Allows for the cooling or warming of the body, which can be crucial for specific procedures like cardioplegia (stopping the heart).

Limitations and Risks

- Requires expert perfusionists for operation and monitoring.
- Risks include:
 - Blood Clotting Issues: Anticoagulants like heparin are required to prevent clots.
 - Air Embolism: Air bubbles can enter circulation despite filters.
 - Inflammatory Response: Exposure of blood to non-biological surfaces can trigger inflammation.
 - Organ Dysfunction: Prolonged bypass can affect kidneys, lungs, or brain.

Clinical Applications

- Coronary Artery Bypass Grafting (CABG)

- Heart Valve Replacement
- Heart Transplants
- Pediatric Cardiac Surgery
- Congenital Heart Defects